

Amazon HackOn Master Preparation Guide

Project: Brand-Based Recommendation + Ad Recommendation + Lost Customer Recovery + Reward Engine

Project Overview

Build a customer intelligence platform that:

- Recommends products based on previous purchases, wishlist, favorite brands and budget.
- Recommends relevant advertisements to users.
- Detects lost customers automatically.
- Sends personalized offers, rewards and recovery campaigns.
- Tracks campaign performance and customer recovery.

Frontend Tech Stack

React
TypeScript
Vite
Tailwind CSS
Shadcn UI
React Router DOM
Axios
Recharts

Backend Tech Stack

Python
FastAPI
SQLAlchemy
Pydantic
Alembic
JWT Authentication
bcrypt
Uvicorn

Database

PostgreSQL

Main Tables:

- customers
- orders
- products
- brands
- wishlists
- recommendations
- campaigns
- rewards
- notifications
- customer_segments

AWS Services

Amazon S3 – Dataset storage, reports and exports
Amazon CloudFront – Frontend delivery
Amazon RDS PostgreSQL – Database
AWS Lambda – Event triggers
Amazon SES – Emails
Amazon SNS – SMS notifications
Amazon CloudWatch – Monitoring and logging
Amazon Cognito (Optional) – Authentication

Recommendation Engine

Libraries:

- Pandas
- NumPy
- Scikit-Learn

Inputs:

- Order History
- Wishlist
- Favorite Brand
- Budget Range

Algorithms:

- Rule-Based Recommendation
- Cosine Similarity
- Collaborative Filtering

Ad Recommendation Engine

Libraries:

- Pandas
- Scikit-Learn

Models:

- Random Forest
- XGBoost

Customer Segmentation

Libraries:

- Pandas
- Scikit-Learn

Algorithm:

- K-Means Clustering

Segments:

- Brand Loyal
- Premium Buyer
- Budget Buyer
- Frequent Buyer
- Inactive Buyer

Lost Customer Recovery Module

Inputs:

- Last Purchase Date

- Order Frequency
- Wishlist Activity

Rule:

No Purchase for 90 Days = Lost Customer

Actions:

- Offer
- Coupon
- Reward
- WhatsApp
- SMS
- Email

Reward Engine

Reward Types:

- Discount Coupons
- Cashback
- Reward Points
- Free Delivery

AI Layer (Supporting Only)

Amazon Bedrock or Groq

Uses:

- Personalized Emails
- Personalized SMS
- Personalized WhatsApp Messages
- Ad Copy Generation

Messaging Services

WhatsApp – Twilio

SMS – Amazon SNS

Email – Amazon SES

Analytics Dashboard

Metrics:

- Recovered Customers
- Lost Customers
- Revenue Recovered
- Ad Performance
- Recommendation CTR
- Campaign Success Rate

Library:

- Recharts

Deployment

Frontend:

- AWS S3
- AWS CloudFront

Backend:

- AWS EC2 or AWS Lambda

Database:

- AWS RDS PostgreSQL

Dataset Structure

Customer Dataset:

customer_id
name
age
preferred_brand
budget_range
wishlist
last_purchase_date
order_frequency

Product Dataset:

product_id
brand
category
price

Development Tools

VS Code – Code Editor

Postman – API Testing

DBeaver – Database Management

Excalidraw – Architecture Diagrams

Canva – PPT Design

AI Tools We Will Use

ChatGPT – Main Development Assistant

Used For:

- Frontend Development
- Backend Development
- Database Design
- AWS Integration
- Recommendation Logic
- Debugging

v0 – Frontend UI Generation

Used For:

- Dashboard UI
- Analytics Pages
- Campaign Pages
- Customer Management Pages

AWS Q Developer

Used For:

- S3
- Lambda
- RDS

- SNS
- SES
- Bedrock

Browsing & Research Tools

Perplexity AI

Used For:

- Latest Documentation
- AWS Examples
- FastAPI Examples
- React Examples
- Library Comparisons

Official Documentation:

- AWS Docs
- FastAPI Docs
- React Docs
- PostgreSQL Docs
- Scikit-Learn Docs

HackOn Day Checklist

- Frontend Boilerplate Ready
- Backend Boilerplate Ready
- AWS Account Ready
- Dataset Ready
- Architecture Diagram Ready
- API Collection Ready
- PPT Template Ready
- Demo Flow Ready
- Database Schema Ready
- Team Tasks Distributed